MIL-B-37074A
22 September 1976
SUPERSEDING
MIL-B-37074
30 December 1974

MILITARY SPECIFICATION

BOTTLE, SAFETY CAP, Glass, Light-Resistant, Prescription

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers light-resistant prescription bottles, with one type of screwcap safety closure suitable for dispensing liquid pharmaceutical preparations.

* 1.2 Classification. Bottles covered by this specification shall be of the following sizes (capacities) expressed in milliliters (ml.) and equivalent fluid ounces (fl. oz.):

Size a - 60 ml. (2 fl. oz.)
Size b - 120 ml. (4 fl. oz.)
Size c - 240 ml. (8 fl. oz.)
Size d - 480 ml. (16 fl. oz.)

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-636 - Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, Attn: Directorate of Medical Materiel, Code ATT, 2800 S. 20th St., Phila., PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6530

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STANDARDS

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM C148 Polariscopic Examination of Glass Containers

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

CONSUMER PRODUCT SAFETY COMMISSION

Poison Packaging Act of 1970 Regulations :

16 CFR 1700.20 Testing Procedure for Special Packaging

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

U. S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Child-Resistance.

* 3.1.1 The contractor shall submit proof to the contracting officer, prior to award of the contract, that the child-resistant bottles to be delivered

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under the contract conform to the Poison Prevention Packaging Act of 1970 (Public Law 91-601 dated December 1970) under 16 CFR 1700.20. Proof shall consist of a copy of data submitted to the Consumer Product Safety Commission on tests conducted in accordance with 16 CFR 1700.20.

3.1.2 Compliance with above paragraph does not excuse compliance with all other requirements of the contract and specifications.

3.1.3 The Government reserves the right to verify that the product furnished to the Government complies with the testing requirements of 16 CFR 1700.20 during the life of the contract and warranty period.

3.2 Material.

3.2.1 Bottles. The bottles shall be fabricated from glass that complies with the provisions of the USP pertaining to light-resistant Type NP glass containers.

3.2.2 Closure caps. The inner and outer caps (see 3.3.2) shall be fabricated from molded plastic material. The material for the outer cap shall be opaque to light.

3.3 Design and construction. The bottles shall comply with the requirements of 3.3.1, and shall be supplied with closure caps on. The closure caps shall comply with the requirements of 3.3.2.

3.3.1 Bottle. The bottle shall have a narrow mouth opening, and shoulders formed by the extension of the neck into the sides of the bottle. The sides of the bottle shall be rectangular or oval, or flat on one side and oval on the other. When tested as specified in 4.4.1, the capacity point of the bottle shall not extend above the bottom level of the neck or below the bottom level of the shoulder. The neck of the bottle shall have an exterior threaded finish to accommodate a screwcap closure (see 3.3.2). The opening and finish of the bottle shall have the following characteristics:

a. A continuous type thread.
b. A flat top surface that is level and free of discernible ridges to make contact with the liner of the closure cap (see 3.3.2.1).

The bottle shall show no strain greater than temper no. 4 when tested as specified in 4.4.2.

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3.3.2 Closure cap. The closure cap shall conform to the requirements of 3.1 and shall be the type that can be opened on a bottle by pushing down on the top while turning. The closure cap shall be a two-piece screw-cap design consisting of an inner cap and an outer cap. The inner cap shall have a continuous type thread for closure on a bottle, and a liner (see 3.3.2.1) to form a seal when full contact is made with the top of the bottle. The outer cap shall meet the requirements of 3.3.2.2, and shall fit over the inner cap in such a manner that the two caps do not separate. Closure on a bottle shall be accomplished by gripping the sides of the outer cap in the same manner as a conventional screwcap and rotating it in a clockwise direction until:

a. The outer cap can no longer be rotated in the clockwise direction.
b. The outer cap rotates freely in the counterclockwise direction without releasing the inner cap from the closure position.

When the closure cap is properly secured to the bottle, the release of the inner cap shall be accomplished by two specific and coordinated actions: pushing down on the top of the outer cap while turning in a counterclockwise direction. The closure cap shall meet the requirements of 3.5.1, 3.5.2 and 3.5.3.

3.3.2.1 Inner cap. A liner shall be secured to the inside of the cap in such a manner that it does not fall out of the cap or is not off-center when the cap is secured to a bottle. The liner shall be supplied with a facing of sufficient thickness and resilience to form an effective seal when full contact is made with the upper edge of the bottle.

* 3.3.2.2 Outer cap. The exterior sides of the outer cap shall have vertical serrations or ridges to facilitate gripping. The manufacturer's normal commercial color for the cap shall be acceptable. Instructions for opening the closure cap shall be imprinted or embossed on the top surface of the cap. The instructions shall be legible. The lettering shall be either a contrasting color to the body of the cap, or not less than 2.5 millimeters (mm.) in height from top to bottom.

3.4 Identification marking. Each bottle shall be permanently and legibly marked with the name or registered trademark of the manufacturer and the rated capacity. The manufacturer's commercial symbol or designation for the milliliter or fluid ounce capacity of his bottle shall be acceptable.

3.5 Bottle assembly.

3.5.1 Tight closure. The bottle with closure cap in place shall be a tight container and shall comply with the test specified in 4.4.3.

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3.5.2 Safety closure durability. The closure cap shall be capable of repeated operations of being secured to and removed from the bottle. When tested as specified in 4.4.4, the closure cap shall function properly without evidence of the following:

a. Deterioration of the components of the closure cap, which impairs the operation and tightness of the cap.
b. Closure cap on bottle able to be opened from safety closure position without applying downward pressure to top of cap.
c. Closure cap not removable from bottle.
d. Components of closure cap become detachable.

3.5.3 Leak-resistant closure. The bottles with closure caps in place shall not leak when tested as specified in 4.4.5.

* 3.6 Workmanship. Workmanship shall be first class throughout. The bottles and closure caps shall be free from defects which detract from their appearance or may impair their serviceability. The bottles and closure caps shall be supplied clean and dry. The bottle shall be free of deformation, cracks, chips, sharp surfaces or sharp edges, and shall be free of rocking and wobbling when placed on a surface that is smooth, flat and level. Mold marks on the bottles shall be acceptable, but shall be free of roughness or sharpness. The closure caps shall be free of sharpness, deterioration, flash, burrs, cracks, chips and breaks. Mold marks on the closure caps shall be acceptable, but shall be free of flash. The bottles shall be free of mold release compound which may be used during the fabrication stage.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

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4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificate of quality. Certificate of quality, supplied by the manufacturer of the glass for the bottles, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results and the specification requirements.

4.2 Sampling.

* 4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I. The unit of product for inspection purposes shall be the individual bottle and closure cap. No more than 10 bottles shall be selected from any unit package (see 5.1.1.1.1).

TABLE I Sampling for examination

	Inspection level	AQL (percent defective)
For visual examination		
Major A defects	II	2.5
Major B defects	II	4.0
Minor defects	II	4.0

4.2.2 For testing. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II. The unit of product for testing purposes shall be the individual bottle and closure cap (where applicable).

TABLE II Sampling for tests

Component (Unit of product)	Characteristic	Requirement	Test procedure	Inspection level
Bottle	Capacity	3.3.1	4.4.1	S-1*
Bottle	Strain	3.3.1	4.4.2	S-2*
Bottle & closure cap	Tight container	3.5.1	4.4.3	**
Bottle & closure cap	Durability of closure	3.5.2	4.4.4	S-2*
Bottle & closure cap	Leakage resistance	3.5.3	4.4.5	S-2*

* Acceptance number shall be zero.
** Sample size and acceptance number shall be as indicated in the test procedure

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* 4.3 Examination. The bottles and closure caps shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with Table III. Examination shall not be restricted to the classified possible defects listed in the table.

TABLE III Classification of defects

Categories	Defects
Major A:	
101	Bottle or closure cap not free of deformation.
102	Bottle or closure cap not free of cracks, chips or breaks.
103	Bottle or closure cap not free of sharp edges or burrs.
104	Bottle not stable when placed on a flat and level surface.
105	Bottle not clean and dry.
106	Thread on closure cap or bottle not free of distortion, stripping or discontinuity.
107	Top of bottle not flat or not free of discernible ridges.
108	Liner not properly aligned or secured within inner screwcap.
109	Closure cap not capable of being properly secured or removed from the bottle.
110	Components of closure cap not properly assembled as a unit or not free of tendency to come apart.
111	Instructions on top of closure cap not legible or complete.
Major B:	
* 151	Bottle not free of mold release compound (e.g. evidence of residue on bottle or on cap liner after cap is assembled to bottle).
Minor	
201	Bottle or closure cap not free of scratches and blemishes.
202	Closure cap not free of imbedded foreign matter.
203	Closure cap not free of flash.

* 4.3.2 Packaging inspection. The inspection of the unit package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

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4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Capacity. A bottle shall be filled with a quantity of water corresponding to the milliliter capacity for the appropriate size indicated in 1.2. The position of the water level in the bottle shall be determined.

4.4.2 Strain. The bottles shall be tested for strain in accordance with ASTM C148, method A or B.

4.4.3 Tight container moisture permeation. The bottles, with closure caps in place, shall be tested in accordance with the procedures for moisture permeation testing included in the USP.

4.4.4 Durability of closure. The closure cap shall be applied to the bottle and tightened with sufficient torque to assure that it is in safety closure position on the bottle. The closure cap shall be released and removed from the bottle by hand in accordance with instructions included on the top of the closure cap. The procedure of tightening the closure cap on the bottle and then removing it from the bottle shall be repeated 25 times. Afterwards, the closure cap shall again be tightened on the bottle. With the closure cap in the safety closure position, it shall be determined whether the closure cap can be removed from the bottle by rotating the outer cap in the counterclockwise direction without applying a downward pressure to the top of the outer cap.

4.4.5. Leakage resistance. The bottle shall be filled with water and the closure cap shall be securely applied, handtight, to at least the manufacturer's recommended application torque. The filled bottle shall be suspended in an inverted position for four hours at a temperature of 25° Centigrade (°C.) ± 5°.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Unit of issue. One package containing the number of bottles specified in table IV constitutes one unit of issue.

5.1.1.1 Level A.

5.1.1.1.1 Unit packaging. The bottles shall be thoroughly clean and screw-cap closures shall be applied. Bottles, in quantities specified in table IV, shall be packaged in a close-fitting box conforming to PPP-B-636, type CF, class domestic. The box shall be provided with full-height partitions extending a minimum of 1/4-inch above the cap of the bottles. The partitions shall form an individual cell for each bottle and shall be fabricated of 0.055 inch solid chip fiber. When bottles are stacked in two or more tiers, a pad fabricated of the same material as the box shall be interposed between each tier. Closure of the box shall be as specified in the appendix of PPP-B-636.

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TABLE IV Packaging quantities

Size	Number of bottles per unit package	Quantity per exterior container
a	72	2 units of issue
b	72	2 units of issue
c	48	2 units of issue
d	24	2 units of issue

5.1.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Two units of issue shall be packed in an exterior container conforming to PPP-B-636, class weather-resistant, grade V3c. Closure, strapping and waterproofing shall be as specified in the appendix of PPP-B-636.

5.2.2 Level B. Two units of issue shall be packed in an exterior container conforming to PPP-B-636, type CF, class domestic. Closure of box shall be as specified in the appendix of PPP-B-636.

5.2.3 Level C. Units shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking. Each unit package and exterior container shall be marked as specified in MIL-STD-129. Marking shall include "Fragile" labels, arrows and the legend: "GLASS -- HANDLE WITH CARE" or similar statement.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Size required (see 1.2).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.2 This specification covers the following items as listed in the Federal Supply Catalog:

National Stock No.	Item Identification
6530-00-111-6351	BOTTLE, SAFETY CAP, Glass, Light-Resistant, 2 oz. (60 cc.), 72s
6530-00-111-6354	BOTTLE, SAFETY CAP, Glass, Light-Resistant, 4 oz. (120 cc.), 72s
6530-00-111-6356	BOTTLE, SAFETY CAP, Glass, Light-Resistant, 8 oz. (240 cc.), 48s
6530-00-111-6359	BOTTLE, SAFETY CAP, Glass, Light-Resistant, 16 oz. (480 cc.), 24s

6.3 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification or those which are commercially available, but is intended to cover the types normally procured to meet Military requirements.

6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DSA-DM

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

MIL-B-37074A - BOTTLE,SAFETY CAP, Glass, Light-Resistant, Prescription

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
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